

Mohit Sharma, Ph.D.

Postdoc Researcher, University of Coimbra, Portugal

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Technical & Research Expertise

Biomaterials innovation, chemical modification, cellulose chemistry, composite formulations, fibre and polymer interactions, sustainable packaging development, barrier coatings, molded cellulose forming, mechanical and optical properties, six-sigma, data analysis, process engineering, paper manufacturing at industrial scale, research and development, analytical techniques, industrial research collaboration, scientific writing, team work, communication, project management and supervision and teaching

Education

Ph.D., Chemical Engineering (PhD program in Biorefineries) <i>University of Coimbra, Portugal</i> ; Final grade: Approved with praise and distinction	2018–2023
M.Tech., Pulp and Paper Engineering <i>Indian Institute of Technology, Roorkee, India</i> ; Final grade: 8.4/10	2012–2014
B.Tech., Biotechnology <i>Vellore Institute of Technology, Vellore, India</i> ; Final grade: 7.7/10	2008–2012

Research Experience

Junior Researcher <i>CERES, Department of Chemical Engineering, University of Coimbra, Portugal</i>	Jul 2023–Present
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- Industrial project *FF2F* – Sustainable Packaging and Products to Replace Fossil Plastic
- Developed bio-based solutions to improve strength and barrier properties of packaging-grade paper.
- Collaborated with commercial and production teams to scale up lab-based innovations for industrial use.
- Conducted pilot trials on nanocellulose production and studied wet-end chemistry to enhance paper strength.
- Prepared detailed technical reports and communicated research outcomes effectively to project partners.
- Presented findings to industrial partners, supporting product development and market launch strategies.
- Participated in customer visits and feedback sessions to ensure R&D efforts are aligned with industry needs.

Doctoral Researcher <i>Department of Chemical Engineering, University of Coimbra, Portugal</i>	Oct 2018–Mar 2023
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- Thesis title: Bio-based coatings for improving paper printing and barrier properties
- Industrial project *Inpactus* R&D to enhance surface properties of printing and writing paper.
- Developed innovative methods to synthesize a starch betainate using a bio-based reagent.
- Prepared bio-based coating formulations to improve the paper printing and barrier properties.
- Presented research findings in quarterly meetings and submitted detailed reports to the industrial partner.

Visiting Scholar <i>Macromolecular and Paper Chemistry, Technical University of Darmstadt, Germany</i>	Sep 2013–Mar 2014
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- Investigated novel photo-reactive polymers to improve wet strength properties of paper.
- Awarded DAAD scholarship to complete the master thesis abroad.

Industrial Experience

Process Engineer <i>Nepa Limited (Recycled fiber-based paper industry), Nepanagar, India</i>	Oct 2014–Sep 2018
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- Led a shift team of 12–14 personnel to ensure consistent newsprint paper production (2 years).
- Played a key role in the revival and mill development plan by supporting the integration of new paper machine and deinking equipment through vendor coordination and early-stage engineering input.
- Maintained regular communication with suppliers for project progress reporting
- Conducted on-site technical inspections at supplier facilities to evaluate compatibility of new equipment.
- Supervised daily installation progress of new equipment, collaborating closely with technical teams.

Post Graduate Engineer Trainee

Aug 2014–Oct 2014

Satia Industries (Pulp and Paper Industry), Muktsar, India

- Participated in hands-on training focused on the operation and maintenance of the Chemical Recovery Plant.
- Gained foundational exposure to pulp production, chemical handling, Evaporators and Recovery Boiler.

Industrial Projects

FF2F – Sustainable Packaging and Products to Replace Fossil Plastic, Portugal (2023–Current)
Inpactus – Innovative Products and Technologies from Eucalyptus, Portugal (2018–2022)
RMDP – Revival & Mill Development Plan, India (2014–2018)

Research & Innovation Projects

- *Ligold* – Lignin-based activated carbon for sustainable recovery of gold from e-waste: a biorefinery approach towards circular economy (2026 Seed Projects for Interdisciplinary Research, Santander Foundation) Jan 2026–Current

Awards & Fellowships

- Project winner within the 7th edition of the seed projects (awarded by Fundação Santander Portugal) – *Ligold project* (as a researcher team member) -funding of 20,000 €.
- PhD scholarship (2018-2022), Portugal – Funded through Project *Inpactus*
- DAAD Scholarship (2013-2014) (Germany)
- MHRD scholarship (2012-2013), awarded by Government of India

Analytical techniques

Spectroscopic techniques (UV-vis, LDS, DLS and FTIR), NMR, SEM, BET analysis, TGA, XRF, contact angle, rheology, elemental analysis, particle charge measurement, zeta potential analysis, fibre morphology analysis, oxygen permeability and WVTR measurements, oil and water absorption tests, contact angle, surface porosity and surface energy analysis, as well as all standard methods for evaluating pulp and paper properties.

Teaching and Pedagogical Experience

I have teaching and mentoring experience in bio-based materials, pulp and paper engineering, and sustainable packaging, gained through doctoral and postdoctoral research at the University of Coimbra (CERES, Department of Chemical Engineering). I have given guest lectures for MSc courses in Pulp and Paper Science, Nanotechnology, and Biorefinery. I have supervised master's students, supporting experimental design, laboratory training, data analysis, and scientific writing. I have also trained students and early-stage researchers within industry-linked R&D projects, particularly in pulp processing, nanocellulose production, additive formulation, and paper characterization. My pedagogical approach is research-driven and student-centered, emphasizing on critical thinking, and the connection between fundamental science and industrial application.

Other academic and Professional Contributions

- Competitive project proposal writing for research funding
- External collaboration with industrial partners and academic research groups
- Reviewed 19 manuscripts for scientific journals such as Carbohydrate Polymers, Cellulose, Food Packaging and Shelf Life, International journal of biological macromolecules, Journal of molecular liquids and others
- Guest Editor of this Special Issue of *Coatings* journal
- Participated in European Researchers' Night (2023, 2025)

Languages

English (Fluent), Portuguese (A2), Hindi (Mother tongue)

Software & Tools

MS Office, Origin, Statistica, TeXstudio

Driving Licence

B, B1 (Portugal)

Personal Information

- Nationality: Indian (Portuguese nationality application submitted on 18 March 2024)
- Current work permit: Portugal, India
- Address (Home): Rua general Humberto Delgado, 105, 3030-327, Coimbra, Portugal
- Address (Work): Department of Chemical Engineering, Rua Silvio Lima, Polo II, 3030-790, Department of Chemical Engineering, Portugal

Patents

- [P1] Sharma, M., Aguado, R., Murtinho, D., Valente, A.J.M., Ferreira, P.J.T., Lourenço, A. (2021). Process for the production of starch betainate. Granted patents in Europe and Portugal; international application filed (WO2022168011).
- [P2] Sharma, M., Valente, A.J.M., Ferreira, P.J.T., Lourenço, A. (2022). Betaine hydrochloride-based formulations for surface treatment of cellulosic substrates. Granted patent in Portugal; pending patents in Europe and Mexico, and an international application filed (WO2024023648).

Scientific articles

- [J1] Sharma, M., Silva A.T., Moura, F., Durães, L., Soares, B., Alves, J.L., Ferreira, P.J.T. (2025). Effect of chitosan-modified industrial white-water on the strength properties of paper containing microfibrillated cellulose. *Cellulose*. DOI: 10.1007/s10570-025-06846-8.
- [J2] Sharma, M., Durães, L., Soares, B., Alves, J.L., Ferreira, P.J.T. (2025). Strengthening of sustainable packaging: impact of microfibrillated cellulose in the presence of industrial white-water. *Cellulose*. DOI: 10.1007/s10570-025-06516-9.
- [J3] Sharma, M., Sharma, M., Esperança, A., Moreira, M.J., Gando-Ferreira, L.M. (2025). Green recovery and optimization of hemicellulose extraction from black liquor using acetic acid and cyclopentyl methyl ether. *Journal of Environmental Chemical Engineering*. DOI: 10.1016/j.jece.2025.116282.
- [J4] Costa, T.B., Matias, P.M.C., Sharma, M., Murtinho, D., Rosa, D.S., Valente, A.J.M. (2024). Recent advances on starch-based adsorbents for heavy metal and emerging pollutant remediation. *Polymers*. DOI: 10.3390/polym17010015.
- [J5] Schnell, C.N., Sharma, M., Pedrosa, J.F.S., Zanuttini, M.A., Mocchiutti, P., Ferreira, P.J.T. (2024). Coatings of xylan/chitosan complexes reinforced with micro/nanocellulose for food packaging applications. *Cellulose*. DOI: 10.1007/s10570-024-06120-3.
- [J6] Sanchez-Salvador, J.L., Rasteiro, M.G., Balea, A., Sharma, M., Pedrosa, J.F.S., Negro, C., Monte, M.C., Blanco, A., Ferreira, P.J.T. (2023). Influence of dispersion of fibrillated cellulose on the reinforcement of coated papers. *International Journal of Biological Macromolecules* DOI: 10.1016/j.ijbiomac.2023.125886.
- [J7] Sharma, M., Aguado, R., Murtinho, D., Valente, A.J.M., Ferreira, P.J.T. (2022). Micro-/nano-fibrillated cellulose-based coating formulations: a solution for improving paper printing quality. *Nanomaterials*. DOI: 10.3390/nano12162853.
- [J8] Sharma, M., Aguado, R., Murtinho, D., Valente, A.J.M., Ferreira, P.J.T. (2021). Synergetic effect of cationic starch (ether/ester) and Pluronic for improving inkjet printing quality of office papers. *Cellulose*. DOI: 10.1007/s10570-021-04206-w.
- [J9] Sharma, M., Aguado, R., Murtinho, D., Valente, A.J.M., Ferreira, P.J.T. (2021). Novel approach on the synthesis of starch betainate by transesterification. *International Journal of Biological Macromolecules*. DOI: 10.1016/j.ijbiomac.2021.05.175.
- [J10] Sharma, M., Aguado, R., Murtinho, D., Valente, A.J.M., Ferreira, P.J.T. (2021). Dynamic penetration profile of starch betainate, Pluronic and PCC coated papers. *KnE Materials Science*, New opportunities for fibrous materials in the ecological transition. DOI: 10.18502/kms.v7i1.11629.
- [J11] Sharma, M., Aguado, R., Murtinho, D., Valente, A.J.M., Mendes De Sousa, A.P., Ferreira, P.J.T. (2020). A review on cationic starch and nanocellulose as paper coating component. *International Journal of Biological Macromolecules*. DOI: 10.1016/j.ijbiomac.2020.06.131.

Conference Presentations

- [C1] Sharma, M., Silva A.T., Moura, F., Soares, B., Alves, J.L., Durães, L., Ferreira, P.J.T. (2025). Impact of industrial white-water on pulp refining and strength properties of microfibrillated cellulose-containing paper

- (Poster presentation). 3rd Iberoamerican Congress on Chemical Engineering (CIBIQ), Lisbon, Portugal.
- [C2] Sharma, M., Silva A.T., Moura, F., Durães, L., Ferreira, P.J.T. (2025). Strength enhancement of LMFC-reinforced packaging paper using chitosan-modified white-water (Poster presentation). 9th International Polysaccharide Conference (EPNOE), Sundsvall, Sweden.
 - [C3] Sharma, M., Soares, B., Durães, L., Ferreira, P.J.T. (2024). Influence of cellulose microfibrils on pulp characteristics in the presence of process white water (Poster presentation). Iberoamerican Congress on Pulp and Paper Research (CIADICYP), Chile.
 - [C4] Sharma, M., Silva, A.T., Soares, B., Durães, L., Ferreira, P.J.T. (2024). Effect of industrial white water on the performance of microfibrillated cellulose to improve paper strength (Poster presentation). Recent Advances in Cellulose Nanotechnology Research, Trondheim, Norway.
 - [C5] Sharma, M., Silva, A., Soares, B., Durães, L., Ferreira, P.J.T. (2024). Effect of lignin-containing cellulose microfibrils on strength properties of packaging grade paper (Poster presentation). 9th International Conference on Bio-based and Biodegradable Polymers (Biopol 2024), Coimbra, Portugal.
 - [C6] Sharma, M., Valente, A.J.M., Ferreira, P.J.T. (2023). On the improvement of paper barrier properties with microfibrillated celluloses and carboxymethyl celluloses-based coating formulations (Poster presentation). International Conference on Nanotechnology for Renewable Materials (TAPPI Nano Division), Canada.
 - [C7] Sharma, M., Feijão, C., Pinto, P., Ferreira, P.J.T. (2023). Effect of microfibrillated cellulose on paper barrier properties for poly(vinyl alcohol)-based coatings (Poster presentation). XXVI International Conference (Tecnicepa) – Forest Biobased Materials, Coimbra, Portugal.
 - [C8] Sharma, M., Feijão, C.M.M., Pinto, P.C.R., Ferreira, P.J.T. (2023). Effect of microfibrillated celluloses on the thermal performance of poly(vinyl alcohol)-based barrier coatings (Poster presentation). 3rd International Workshop on Biorefinery of Lignocellulosic Materials (IWBLCM-2023), Córdoba, Spain.
 - [C9] Sharma, M., Valente, A.J.M., Ferreira, P.J.T. (2022). Rheological properties of micro-/nanofibrillated cellulose-based coating formulations (Poster presentation). WCPT-9, Madrid, Spain.
 - [C10] Sharma, M., Aguado, R., Murtinho, D., Valente, A.J.M., Ferreira, P.J.T. (2022). New developments in paper coating with micro-/nanofibrillated cellulose for improving printing quality (Oral presentation). Iberoamerican Congress on Pulp and Paper Research (CIADICYP), Girona, Spain.
 - [C11] Sharma, M., Aguado, R., Valente, A.J.M., Murtinho, D., Ferreira, P.J.T. (2021). Dynamic penetration profile of starch betainate, Pluronic and PCC coated papers (Poster). 1st FibEnTech International Conference – New Opportunities for Fibrous Materials, Portugal.
 - [C12] Sharma, M., Aguado, R., Murtinho, D., Valente, A.J.M., Ferreira, P.J.T. (2021). Ball milling: a new development in the synthesis of cationic starch ester (Oral presentation). 7th International Polysaccharide Conference (EPNOE), Nantes, France.
 - [C13] Sharma, M., Aguado, R., Murtinho, D., Valente, A.J.M., Mendes de Sousa, A.P., Ferreira, P.J.T. (2021). Effect of enzymatic pretreatment on starch cationization (Oral presentation). XXV International Forest, Pulp & Paper Conference (Tecnicepa), Portugal.